$\min f(x) \quad \text{s.t.} \quad Ax = b \quad (P)$

$L(x, \lambda) = f(x) + \langle \lambda, Ax - b \rangle$

$g(\lambda) = \min_x L(x, \lambda)$

$\max g(\lambda) \quad \text{s.t.} \quad \lambda \geq 0 \quad (D)$

Dual Aes

$\downarrow$ $L + \frac{1}{2}\rho h^2$ 增广

ALM

$\downarrow$ 当 $\min f + g$ 时

ADMM

Dual Ascent $\quad x^{k+1} = \operatorname{argmin} L(x, \lambda^k)$

$$\lambda^{k+1} = \lambda^k + \alpha(Ax^{k+1} - b)$$

Augment Lagrangian $\quad L_\rho := f(x) + \langle \lambda, Ax - b \rangle + \frac{\rho}{2} \| Ax - b \|^2$

$$x^{k+1} = \operatorname{argmin} L_\rho(x, \lambda^k)$$

$$\lambda^{k+1} = \lambda^k + \rho(Ax^{k+1} - b)$$

ADMM (Alternating Direction Method of Multipliers)

$\min_{x,z} f(x) + g(z) \qquad$ convex

$\text{s.t.} \quad Ax + Bz = c \qquad$ Linear Constraint

$$L_\rho := f(x) + g(z) + \langle \lambda, Ax + Bz - c \rangle + \frac{\rho}{2} \| Ax + Bz - c \|^2$$

$$= f(x) + g(z) + \frac{\rho}{2} \left[\| Ax + Bz - c + \frac{1}{\rho}\lambda \|^2 - \| \frac{1}{\rho}\lambda \|^2 \right]$$

$$= f(x) + g(z) + \frac{\rho}{2} \left[\| Ax + Bz - c + u \|^2 - \| u \|^2 \right]$$

Scaled $u := \frac{1}{\rho}\lambda$

ADMM Iterative

$$\left.\begin{aligned} x^{k+1} &= \operatorname{argmin} L_\rho(x, z^k, \lambda^k) \\ z^{k+1} &= \operatorname{argmin} L_\rho(x^{k+1}, z, \lambda^k) \end{aligned}\right\} \text{Primal}$$

$$\lambda^{k+1} = \lambda^k + \rho(Ax^{k+1} + Bz^{k+1} - c) \quad \Big\} \text{Dual}$$

$$u^{k+1} = u^k + Ax^{k+1} + Bz^{k+1} - c$$

Opt Condition (iff):

$$Ax^* + Bz^* = c \qquad \nabla_x L = \nabla f^* + A^T\lambda^* = 0$$

$$\nabla_z L = \nabla g^* + B^T\lambda^* = 0$$

Residual $r^k = \lambda^k - \lambda^{k-1} = Ax^k + Bz^k - c$

Dual Residual $s^k = \rho A^T B(z^{k+1} - z^k) = \nabla f^{k+1} + A^T\lambda^{k+1}$

Pf: $x^{k+1} = \operatorname{argmin} L_\rho(x, z^k, \lambda^k)$

$$0 = \nabla f^{k+1} + A^T\lambda^k + \rho A^T(Ax^{k+1} + Bz^k - c)$$

$$= \nabla f^{k+1} + A^T[\lambda^k + \rho(Ax^{k+1} + Bz^{k+1} - c)] - \rho A^T B(z^{k+1} - z^k)$$

$$\rho A^T B(z^{k+1} - z^k) = \nabla f^{k+1} + A^T\lambda^{k+1}$$

收敛证明中能证2个残差收敛 $r^k, s^k \to 0$

即 $\lambda^{k+1} - \lambda^k \to 0$, $z^{k+1} - z^k \to 0$,

之后再证序列收敛 $z^k \to z^*$, $\lambda^k \to \lambda^*$

e.g. $\min f(x)$ s.t. $x \in C$

$$\min f(x) + I_C(z) \quad \text{s.t. } x - z = 0$$

$$L_\rho(x, z) = f(x) + I_C(z) + \langle \lambda, x - z \rangle + \frac{\rho}{2}\|x - z\|^2$$

$$x^{k+1} = \operatorname{argmin} f(x) + \frac{\rho}{2}\|x - z^k + u^k\|$$

$$z^{k+1} = \operatorname{Proj}_C(x^{k+1} + u^k)$$

$$u^{k+1} = u^k + x^{k+1} - z^{k+1}$$

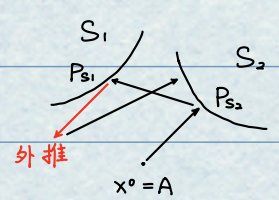
e.g. AP - Dykstra Find $x \in S_1 \cap S_2$

$$\min I_{S_1}(x) + I_{S_2}(z) \quad \text{s.t. } x = z$$

$$x^{k+1} = \operatorname{Proj}_{S_1}(z^k - u^k)$$

$$z^{k+1} = \operatorname{Proj}_{S_1}(x^{k+1} + u^k)$$

$$u^{k+1} = u^k + x^{k+1} - z^{k+1}$$



ADMM for CS Algo

BP - ADMM $\quad \min \|x\|_1 \quad \text{s.t.} \quad y = Ax$

$$\min \|x\|_1 + I_C(z) \quad \text{s.t.} \quad x = z \qquad C = \{x \mid y = Ax\}$$

$$L_\rho(x, z) = \|x\|_1 + I_C(z) + \langle \lambda, x - z \rangle + \frac{\rho}{2}\|x - z\|^2$$

$$x^{k+1} = \operatorname{argmin} \|x\|_1 + \langle \lambda^k, x - z^k \rangle + \frac{\rho}{2}\|x - z^k\|^2$$

$$= S_{\frac{1}{\rho}}(z^k + u^k)$$

$$z^{k+1} = \operatorname{argmin} I_C(z) + \langle \lambda^k, x^{k+1} - z \rangle + \frac{\rho}{2}\|x^{k+1} - z\|^2$$

$$= \operatorname{argmin} \|z - x^{k+1} + \tfrac{1}{\rho}\lambda^k\| \quad \text{s.t.} \quad y = Az$$

$$= \operatorname{Proj}_C(x^{k+1} - u^k) = \operatorname{Proj}_A^{\perp}(x^{k+1} - u^k) + A^{\dagger} y$$

$$u^{k+1} = u^k + x^{k+1} - z^{k+1}$$

LASSO - ADMM $\quad \min \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|x\|_1$

$$\min \frac{1}{2}\|y - Ax\|^2 + \lambda\|x\|_1$$

$$\min \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|z\|_1 \quad \text{s.t.} \quad x = z$$

$$L_\rho(x, z) = \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|z\|_1 + \langle \lambda, x - z \rangle + \frac{\rho}{2}\|x - z\|^2$$

$$x^{k+1} = \operatorname{argmin} \frac{1}{2}\|y - Ax\|_2^2 + \langle \lambda^k, x^{k+1} - z \rangle + \frac{\rho}{2}\|x^{k+1} - z\|^2$$

$$= (A^T A + \rho I)^{-1}(A^H y + \rho z^k - \lambda^k)$$

$$z^{k+1} = S_{\lambda/\rho}(x^{k+1} - u^k)$$

$$u^{k+1} = u^k + x^{k+1} - z^{k+1}$$